

BloodLink — Sample Bug Reports

Project: BloodLink v2.0 — Blood Bank Management & Intelligent Donor Matching System

Document Type: QA Artifact — Sample Bug Reports

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Total Bugs Filed: 8

Bug Summary Table

Bug ID	Title	Module	Severity	Priority	Status
BL-BUG-001	Login accepts empty password via Enter key	Authentication	Critical	P1	New
BL-BUG-002	Inventory allows adding negative blood units	Inventory	High	P1	New
BL-BUG-003	Matching engine returns 0 donors for AB-	Emergency Matching	Critical	P1	Open
BL-BUG-004	"Notify All" button remains disabled (Safari)	Emergencies	Medium	P2	New
BL-BUG-005	timeAgo() shows negative values for future dates	Dashboard	Low	P3	New
BL-BUG-006	Delete modal closes on body click	Inventory	Medium	P3	New
BL-BUG-007	Notification response resets on page refresh	Notifications	High	P2	Open

Bug ID	Title	Module	Severity	Priority	Status
BL-BUG-008	Dashboard inventory bar has hardcoded scale	Dashboard	 Low	P4	New

BL-BUG-001 — Login accepts empty password field when pressing Enter key

Field	Value
Bug ID	BL-BUG-001
Reporter	QA Team — Salma Nour
Date Reported	2026-04-28
Module	Authentication (Login Page)
Severity	 Critical
Priority	P1 (Immediate)
Status	New
Assigned To	Backend Team
Environment	Windows 11, Chrome 126, 1920×1080

Description

When a user enters a valid email address and leaves the password field empty, pressing the Enter key bypasses the HTML `required` attribute and submits the form. The backend responds with an authentication error, but the frontend shows a generic "Invalid credentials" message instead of a field-level validation error.

Steps to Reproduce

1. Navigate to `/login`.
2. Enter `admin@bloodlink.org` in the email field.
3. Leave the password field completely empty.
4. Press the **Enter** key (instead of clicking the "Sign In" button).

Expected Result

- The browser should block form submission with a native "Please fill out this field" tooltip on the password input.
- No API call should be made.

Actual Result

- The form submits to the backend.
- A red error banner appears: "Invalid email or password."
- Console shows `POST /auth/login 401 Unauthorized`.

Evidence

```
Console Output:
POST http://localhost:3001/auth/login 401 (Unauthorized)
Response: {error: "Invalid credentials"}
```

Additional Context

- This occurs only when pressing Enter; clicking the Sign In button correctly triggers HTML5 validation.
- Frequency: **Always reproducible**.
- Root cause: Potential browser-specific behavior where `required` validation is skipped on implicit form submission.

BL-BUG-002 — Inventory allows adding negative blood units

Field	Value
Bug ID	BL-BUG-002
Reporter	QA Team — Ahmed Mostafa
Date Reported	2026-04-29
Module	Blood Inventory Management
Severity	High

Field	Value
Priority	P1 (Immediate)
Status	New
Assigned To	Frontend Team
Environment	Windows 11, Firefox 127, 1536×864

Description

The "Add Blood Units" modal accepts negative numbers in the "Units Available" field. Submitting the form with a negative value (e.g., `-50`) successfully adds a record with negative inventory, corrupting aggregate statistics on the dashboard.

Steps to Reproduce

1. Log in as `admin@bloodlink.org`.
2. Navigate to `/inventory`.
3. Click **"Add Blood Units"** button.
4. Select blood type **O+**.
5. Enter **-50** in the "Units Available" field.
6. Click **"Add to Inventory"**.

Expected Result

- The form should reject negative values.
- An inline validation error should display: "Units must be a positive number."

Actual Result

- The record is created successfully with `-50` units.
- The dashboard's "Total Inventory" card shows a reduced total.
- The status is computed as "Critical" due to the negative value passing through `getStatusFromUnits()`.

Evidence

Inventory Table Row After Submission:

Blood Type: O+ | Units: -50 | Status: Critical | Trend: +0%

Additional Context

- Root cause: The `<input type="number" min="0">` attribute is present but `parseInt(formUnits)` does not validate the parsed value before calling `addInventoryItem()`. The backend also lacks server-side validation for negative units.
- Frequency: **Always reproducible.**
- Suggested fix: Add `if (units <= 0) { toast("error", ...); return; }` before the API call.

BL-BUG-003 — Emergency donor matching returns 0 donors for AB- despite available AB- donors

Field	Value
Bug ID	BL-BUG-003
Reporter	QA Team — Dr. Hana Mostafa
Date Reported	2026-04-29
Module	Emergency Matching Engine
Severity	 Critical
Priority	P1 (Immediate)
Status	Open
Assigned To	Backend Team
Environment	Windows 11, Chrome 126, 1920×1080, Backend v1.0.0

Description

When creating an emergency request for **AB-** blood, the Matching Engine reports "0 found" compatible donors, even though there is an eligible, available AB- donor (Omar Abdelrahman, donor ID `d-5`) in the system. The matching algorithm appears to filter donors by `available === true` **and** `eligible === true`, but Omar's `lastDonation` was over 4 months ago and all eligibility flags are correctly set.

Steps to Reproduce

1. Log in as `admin@bloodlink.org`.
2. Navigate to `/emergencies`.
3. Click **+** to create a new emergency request.
4. Set: Hospital = "Cairo General", Department = "ICU", Blood Type = **AB-**, Units = 2, Urgency = **Critical**.
5. Submit the form.
6. Observe the Matching Engine panel on the right.

Expected Result

- Compatible Donors: **1 found** (Omar Abdelrahman, AB-, reliability 99%).
- The donor card should appear in the "Top Recommended Donors" list.

Actual Result

- Compatible Donors: **0 found**.
- "No compatible donors found." message is displayed.

Evidence

Matching Engine Panel:

Required Type: AB- Compatible Donors: 0 found High Reliability: 0 match

GET `/api/emergencies/:id/match` → Response:

```
{ totalCompatible: 0, highReliability: 0, donors: [] }
```

Additional Context

- Suspected root cause: The compatibility engine may not include AB- ↔ AB- exact match; it may only implement the universal donor table (O- as universal donor) but miss same-type matches.
- This is **life-threatening** in a production scenario.
- Frequency: **Always reproducible**.

BL-BUG-004 — "Notify All" button remains disabled after successful notification

Field	Value
Bug ID	BL-BUG-004
Reporter	QA Team — Ahmed Mostafa
Date Reported	2026-04-30
Module	Emergency Notifications
Severity	● Medium
Priority	P2 (Next Sprint)
Status	New
Assigned To	Frontend Team
Environment	macOS Sonoma, Safari 17.5, 1440×900

Description

After clicking "Notify All Top Matches" on the Emergencies page, the button shows a loading spinner and then the success toast appears. However, the button remains in a visually disabled state (`opacity-60` , `cursor-wait`) even though `notifyingAll` has been set back to `false` . The button becomes functional again only after switching to a different emergency and back.

Steps to Reproduce

1. Log in as `admin@bloodlink.org` .
2. Navigate to `/emergencies` .
3. Select an emergency with matched donors.
4. Click **"Notify All Top Matches"**.
5. Wait for the success toast to appear.
6. Observe the button state.

Expected Result

- The button should return to its normal active styling.
- Ideally, the button text should change to "All Donors Notified ✓" with a green accent.

Actual Result

- The button appears grayed out and unclickable for the current session.
- The button text reverts to "Notify All Top Matches" but the styling remains disabled.

Evidence

```
Button element after notification (inspected via DevTools):  
<button disabled class="...disabled:opacity-60 disabled:cursor-wait">  
  Notify All Top Matches  
</button>
```

Additional Context

- Suspected cause: Safari's rendering engine caches the `disabled` attribute state. The React state updates correctly, but Safari does not re-paint the button styles.
- Frequency: **Always on Safari, not reproducible on Chrome/Firefox.**

BL-BUG-005 — Donor dashboard "time ago" shows negative values for future-dated notifications

Field	Value
Bug ID	BL-BUG-005
Reporter	QA Team — Salma Nour
Date Reported	2026-04-30
Module	Dashboard — Donor View
Severity	 Low
Priority	P3 (Backlog)
Status	New
Assigned To	Frontend Team
Environment	Windows 11, Chrome 126, 1920×1080

Description

The `timeAgo()` utility function does not handle notification timestamps that are in the future (e.g., scheduled maintenance windows). When a notification has a future timestamp, the function computes a negative `diff`, resulting in displays like "-120m ago" or "NaN d ago".

Steps to Reproduce

1. Log in as a donor account (or observe from admin dashboard).
2. Check the notification with title "**System Maintenance**" (timestamp: `2026-04-05 14:00`).
3. If the system clock is before this timestamp, observe the time display.

Expected Result

- Future-dated notifications should display "Upcoming" or "Scheduled" instead of a negative relative time.

Actual Result

- The notification displays "**-345600m ago**" or similar negative values.

Evidence

Notification card footer text:
"System Maintenance • -5760m ago"

Source: `page.tsx` lines 20-27:

```
function timeAgo(dateStr) {  
  const diff = Date.now() - new Date(dateStr).getTime();  
  const mins = Math.floor(diff / 60000); // ← no guard for negative  
  ...  
}
```

Additional Context

- Frequency: **Conditionally reproducible** (depends on system clock vs. notification timestamp).
 - Suggested fix: Add `if (diff < 0) return "Upcoming";` at the top of `timeAgo()` .
-

BL-BUG-006 — Inventory delete confirmation modal closes when clicking inside modal body

Field	Value
Bug ID	BL-BUG-006
Reporter	QA Team — Dr. Hana Mostafa
Date Reported	2026-04-30
Module	Blood Inventory — Delete Action
Severity	● Medium
Priority	P3 (Backlog)
Status	New
Assigned To	Frontend Team
Environment	Windows 11, Edge 126, 1920×1080

Description

The delete confirmation modal on the Inventory page uses `onClick={() => setDeletingId(null)}` on the backdrop overlay, which correctly dismisses the modal when clicking outside. However, the inner modal `<div>` does **not** include `e.stopPropagation()`, unlike the Add/Edit modal which does. As a result, clicking anywhere inside the delete modal (including the text or whitespace) also dismisses it.

Steps to Reproduce

1. Log in as `admin@bloodlink.org`.
2. Navigate to `/inventory`.
3. Click the `:` actions menu on any blood type row.
4. Click **Delete**.
5. In the confirmation dialog, click on the text "Are you sure you want to remove..." (not on any button).

Expected Result

- Clicking inside the modal body should not close it.

- Only clicking "Cancel" or the backdrop should dismiss the modal.

Actual Result

- The modal closes immediately, as if the user clicked the backdrop.

Evidence

```
inventory/page.tsx line ~352:
// Delete modal – missing e.stopPropagation():
<div className="bg-panel ..." <← no onClick handler
  <h2>Confirm Deletion</h2>
  ...
</div>

// Compare with Add/Edit modal at line ~297:
<div className="bg-panel ..." onClick={(e) => e.stopPropagation()} <← has handler
```

Additional Context

- Frequency: **Always reproducible.**
- Suggested fix: Add `onClick={(e) => e.stopPropagation()}` to the inner `<div>` of the delete confirmation modal.

BL-BUG-007 — Notification "response" state resets to "pending" on page refresh

Field	Value
Bug ID	BL-BUG-007
Reporter	QA Team — Salma Nour
Date Reported	2026-05-01
Module	Donor Notifications / State Management
Severity	● High
Priority	P2 (Next Sprint)
Status	Open

Field	Value
Assigned To	Full Stack Team
Environment	Windows 11, Chrome 126, 1920×1080

Description

When a donor accepts or declines an emergency request, the notification card correctly updates to show "Accepted ✓" or "Declined X". However, refreshing the page (F5) resets all notification responses back to "pending", causing the Accept/Decline buttons to reappear. The donor's response is not persisted to the backend or local storage.

Steps to Reproduce

1. Log in as a donor user.
2. On the dashboard, find an emergency notification with "Accept" / "Decline" buttons.
3. Click **Accept**.
4. Observe the card updates to "Accepted ✓" with a green badge.
5. Press **F5** to refresh the page.

Expected Result

- The notification should remain in the "Accepted" state after refresh.
- The Accept/Decline buttons should not reappear.

Actual Result

- All notifications revert to `response: "pending"`.
- The donor can accept the same notification again, potentially sending duplicate confirmations to the hospital.

Evidence

Before refresh: Emergency notification shows ✓ Accepted badge

After refresh: Same notification shows [Accept] [Decline] buttons

Root cause in context.tsx:

`respondToNotification()` updates React state in-memory only.

`data.ts` reinitializes with hardcoded `response: "pending"` on every page load.

Additional Context

- Root cause: The `respondToNotification` function in `context.tsx` updates React state in memory but the mock data in `data.ts` reinitializes with hardcoded `response: "pending"` on every page load.
- In a production environment, this should persist via a `PATCH /notifications/:id` API call.
- Frequency: **Always reproducible.**

BL-BUG-008 — Dashboard inventory bar uses hardcoded denominator of 500

Field	Value
Bug ID	BL-BUG-008
Reporter	QA Team — Ahmed Mostafa
Date Reported	2026-05-01
Module	Admin Dashboard — Live Inventory Status
Severity	● Low
Priority	P4 (Nice-to-have)
Status	New
Assigned To	Frontend Team
Environment	Windows 11, Chrome 126, 1366×768

Description

On the admin dashboard, the "Live Inventory Status" section renders a horizontal progress bar for each blood type. The bar width is calculated as `(item.units / 500) * 100`, capped at 100% via `Math.min()`. While the cap works correctly, the hardcoded denominator of `500` means that any blood type with stock significantly above 500 units will always show a full 100% bar, providing no visual differentiation between 500 units and 2000 units.

Steps to Reproduce

- 1. Log in as `admin@bloodlink.org`.
- 2. Navigate to the main dashboard `/`.
- 3. Observe the "Live Inventory Status" section.
- 4. Note that **O+** (420 units) shows a nearly full bar, and if a type had 800 units, it would show the exact same full bar as 500 units.

Expected Result

- The bar width denominator should be dynamically calculated based on the maximum units across all blood types, so that the bars provide meaningful visual comparison.

Actual Result

- All blood types with ≥ 500 units display an identical 100% bar width regardless of actual count.

Evidence

Bar comparison (visual approximation):

O+ 420U (84% bar)

A+ 310U (62% bar)

AB+ 500U (100% bar) ← same as 2000U

Source: page.tsx line 133:

style={{ width: `\${Math.min((item.units / 500) \* 100, 100)}%` }}

Additional Context

- Suggested fix: Replace `500` with `Math.max(...inventory.map(i => i.units))` for a dynamic scale.
- Frequency: **Always reproducible.**

Appendix A: Severity & Priority Distribution

Severity	Priority	Count	Bug IDs
Critical	P1	2	BL-BUG-001, BL-BUG-003
High	P1-P2	2	BL-BUG-002, BL-BUG-007

Severity	Priority	Count	Bug IDs
● Medium	P2–P3	2	BL-BUG-004, BL-BUG-006
● Low	P3–P4	2	BL-BUG-005, BL-BUG-008

Appendix B: Module Coverage Matrix

Module	Bugs Filed	IDs
Authentication	1	BL-BUG-001
Blood Inventory	2	BL-BUG-002, BL-BUG-006
Emergency Matching	1	BL-BUG-003
Notifications / Emergencies	2	BL-BUG-004, BL-BUG-007
Dashboard	2	BL-BUG-005, BL-BUG-008